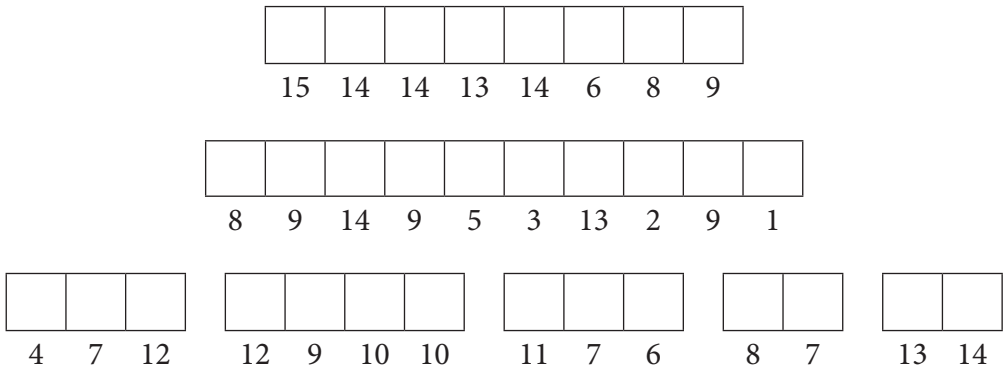
Name: _____

WORKSHEET

Simultaneous equations

Ability is what you are capable of doing. Motivation determines what you do …



Lou Holtz

Questions

Solve the following simultaneous equations by substitution or elimination.

1 $x + y = 5$
$x - y = 1$

2 $x + y = 7$
$x - 2y = -8$

3 $x + y = -2$
$x - y = 4$

4 $2x - y = 6$
$x + 2y = 8$

5 $x + 3y = -1$
$4x + y = 18$

6 $3x + 2y = 5$
$2x - 4y = -18$

7 $x + 5y = 1$
$3x - 2y = -14$

8 $2x - 7y = 17$
$5x + 3y = -19$

9 $4x - 5y = 7$
$3x + 2y = 34$

10 $8x - 3y = -74$
$3x + 6y = 15$

11 CAS $7x + 2y = 3$
$4x + 5y = -33$

12 CAS $x + 2y + z = 7$
$x + y + 3z = 12$
$x - y + z = 4$

13 CAS $2x + y - z = -4$
$x + 3y + z = 11$
$-x - y + 4z = 7$

14 CAS $5x + y + 2z = 8$
$x + 3y - 4z = -30$
$2x - 4y + z = 26$

15 CAS $3x - y + z = 15$
$x + 5y - 2z = -12$
$-4x + 2y + 3z = 1$

Solutions

A $x = 3, y = -1, z = 5$

D $x = -2, y = -3$

E $x = 8, y = 5$

H $x = 4, y = 2$

I $x = -3, y = 4, z = 2$

L $x = -7, y = 6$

M $x = 1, y = -3$

N $x = 2, y = 5$

O $x = -4, y = 1$

R $x = 5, y = -2$

S $x = 3, y = 2$

T $x = 1, y = -5, z = 4$

U $x = -1, y = 4$

W $x = 2, y = 1, z = 3$

Y $x = 3, y = -9$

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